# Gabriel Koerich | Software Engineer

## Summary

Senior software engineer and entrepreneur with 15+ years of professional experience, coding since age 13. I build backends that do many things at once: queues, background workers, retries and concurrency. I have written Rust since 2021. Founded Bulldesk (acquired 2021), a marketing automation and CRM platform that ran 500M+ background jobs a month. Currently building Solana DeFi at Lulo, and AI agent tooling around Claude Code, Codex and OpenCode. BBA in Finance.

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## Core Competencies

**Distributed Systems & Concurrency:** Event-driven architecture, queue-driven backends, background workers, retries, idempotency, async Rust with Tokio (worker pools, backpressure), 500M+ background jobs a month at Bulldesk

**AI & Agent Orchestration:** Claude Code, Codex, OpenCode in daily production use. Multi-agent orchestration (orch), automating engineering workflows, prompt engineering. Available for consulting.

**Blockchain:** Solana (Rust, Anchor), EVM (Solidity, Foundry, Hardhat), DeFi integrations, onchain and off-chain indexing, testing (unit, fuzz, integration)

**Languages:** Rust, TypeScript, Python

**Data & Infrastructure:** PostgreSQL, MySQL, Redis, Beanstalkd, SQS, Docker, GitHub Actions, Terraform, AWS, GCP

**Frontend:** Vue.js, React, Vite

## Professional Experience

### Senior Software Engineer, Lulo (Nov 2022 – Present)

Smart contract development for the DeFi protocol. Designed two pools: a security pool that protects users against protocol hacks while earning yield, and a higher-yield pool with standard DeFi risk.

- Architected the protocol and its CPI integrations with major Solana protocols
- Built SDKs, CLIs and test suites (Rust unit and Anchor integration)
- Developed EVM protocol integrations using Solidity and Forge

**Stack:** Rust, Solana, Anchor, TypeScript, GitHub Actions, GCP

### Solana Smart Contract Engineer, Freelance (Oct 2021 – Nov 2022)

Built three Solana projects: an NFT marketplace with custom auctions, a price prediction game with an onchain oracle, and DeFi contracts using Orca Whirlpools. Each shipped with tests and TypeScript and Rust SDKs.

**Stack:** Rust, Solana, Anchor, TypeScript

### Senior Software Engineer, Covie (May 2021 – Oct 2021)

Built scrapers and browser automation that collected policy and claims data from insurance company portals at high volume.

**Stack:** Symfony, Node, PostgreSQL

**Founder, Bulldesk (Sep 2015 – May 2021)**

Founded and scaled the first Brazilian platform to combine marketing automation and CRM. Built core architecture, led a team of 4 developers, managed infrastructure, and ran product development and company finances. Acquired in 2021.

- Scale: 80M+ monthly user requests, 500M+ monthly background jobs
- Event-driven, queue-driven backend: automation engine, campaigns and email on background workers
- Load-balanced app servers, dedicated worker servers, MySQL replication, Beanstalkd, Redis and SQS queues
- Automation engine processing 1M+ leads through multi-step campaigns

**Stack:** Laravel, Node, MySQL, Beanstalkd, Redis, SQS, Vue.js, AWS

## Education

- **BBA Finance**: Universidade do Sul de Santa Catarina
- **English Language Studies**: LSI, London

**Languages:** Portuguese (native), English (fluent)